

Mobile Device Digital Forensics Investigation

MicroSD Image, GPS, Communications and Timeline Analysis

Raliegth Barnett | Cybersecurity & Networking Portfolio

Portfolio Case Study - Detailed Project Oversight and Completion Record

All offensive-security activities described in this document were performed in authorized, isolated academic laboratory environments.

1. Project Overview

An academic mobile-forensics case study examining an E01/E02 MicroSD image with FTK Imager and Autopsy. The project verified evidence integrity and correlated GPS, Maps history, SMS, call logs, contacts, installed applications, and search history into a documented timeline.

Project Field	Details
Role and Responsibility	Mobile forensic examiner, artifact analyst, timeline analyst, and report author.
Environment and Tools	FTK Imager, Autopsy 4.22.1, Android/MicroSD E01/E02 evidence segments.
Completion Status	Completed - documented, tested, and included in the master portfolio.
Portfolio Classification	Academic / personal technical project with recruiter-facing case-study documentation.

2. Project Objectives and Success Criteria

- Verify image integrity before analysis.
- Identify device and installed-application artifacts.
- Recover and interpret GPS and Google Maps history.
- Analyze SMS, call logs, and contacts.
- Review application search history and travel-related indicators.
- Construct a timeline and state limitations.

3. Project Oversight and Execution Process

3.1 Planning and Scope

Defined the authorized environment, project goal, tools, expected outputs, and safety limitations.

3.2 Implementation

Executed the technical tasks and maintained notes, commands, screenshots, and configuration evidence.

3.3 Validation and Testing

Compared actual output with objectives, confirmed repeatable behavior, and collected proof of completion.

3.4 Analysis and Reporting

Converted technical evidence into findings, defensive interpretation, limitations, and recommendations.

3.5 Portfolio Finalization

Prepared a recruiter-friendly summary, detailed case study, screenshots, source report, and public-release guidance.

4. Technical Methodology

1. Verified recorded MD5 and SHA-1 values.
2. Ran Autopsy ingest and reviewed system and account artifacts.
3. Recovered cached GPS coordinates and destination history.
4. Examined SQLite-derived SMS, call, and contact records.
5. Reviewed application-store search history.
6. Correlated independent artifacts into a time-ordered narrative.
7. Separated direct device evidence from investigative inference.

5. Deliverables and Completion Evidence

Deliverable	Status	Evidence
Technical implementation / lab execution	Complete	Commands, configurations, code, or forensic workflow documented.
Testing and validation	Complete	Results and screenshots included below and in the original report.
Detailed project documentation	Complete	This case-study document and README.
Source/original evidence	Complete	Original report or source files stored in this project folder.
Recruiter-ready description	Complete	Summary and skills section included.
Public publishing review	Required before posting	Remove credentials, personal identifiers, confidential case data, and unnecessary target details.

6. Results and Outcomes

- Multiple independent artifact categories supported a consistent geographic and identity narrative.
- The report documented evidence integrity, location history, communications, contacts, applications, and a reconstructed timeline.
- The project demonstrates the importance of corroboration across separate mobile databases.

7. Challenges, Decisions, and Lessons Learned

- Time-zone normalization and interpretation of cached versus user-initiated location records.
- Avoiding attribution of device activity to a person without corroborating evidence.

8. Screenshots and Supporting Evidence

Both MD5 and SHA1 hashes were verified against the acquisition manifest, confirming the forensic image is authentic and unmodified. Chain of custody is intact and the evidence is admissible.

3. Device & System Information

Autopsy's automated ingest identified the following system-level artifacts on the Noonien.E01 data source:

Operating System Information	1 artifact recovered (Android-based mobile OS consistent with the associated Samsung Galaxy SIII device)
Installed Programs	53 installed applications recovered
Communication Accounts	106 accounts recovered (2 device-level, 27 phone, 77 other)

4. GPS & Location Evidence

Two categories of location data were recovered from the image by Autopsy's Geolocation module: a cached browser geoposition bookmark and a Google Maps navigation route with two timestamped entries.

4.1 GPS Bookmark — CachedGeoposition.db

Source	CachedGeoposition.db (Browser Location History)
Latitude	42.6801053
Longitude	-83.1355078
Date/Time	2013-12-30 14:31:33 COT (Autopsy case timezone)
Geocoded Area	Oakland County, Michigan, USA — Detroit metropolitan area (vicinity of Madison Heights / Royal Oak, approximately 8 km east of Southfield)

4.2 GPS Route — Google Maps Navigation History

Two navigation entries were recovered from the Android destination history (da_destination_history), both produced by Google Maps History:

Date/Time (COT)	Coordinates	Destination
2013-08-09 13:23:06	42.50494, -83.316652	29555 Northwestern Hwy, Southfield, MI 48034
2013-08-09 13:31:14	(second Google Maps History entry)	Same session / same destination area

The Google Maps navigation target (29555 Northwestern Highway, Southfield, MI 48034) is a real commercial address corresponding to La Mirage Plaza, a retail center in Southfield anchored

Evidence integrity, device information, and GPS sections.

2014-05-20 (morning)	SMS thread with +1 (248) 929-4645 regarding a family hospital situation.
2014-05-20 (midday)	Automated customer-service SMS addressed to "Peter" (identity corroboration).
2014-05-21	1,500 call log entries concentrated on this date, including an outgoing call to +1 (860) 759-2990.

Note on timezone: Autopsy is configured to display timestamps in COT (Colombia Time, UTC-5). This is numerically equivalent to US Eastern Standard Time for the winter entries but one hour behind EDT for the summer entries. Times should be re-normalised to America/Detroit before presentation in court.

8. Findings & Conclusions

1. SUSPECT LOCATION: GPS Route and CachedGeoposition data place the device in Oakland County, Michigan, with a specific Google Maps navigation target at 29555 Northwestern Hwy, Southfield, MI 48034 on 2013-08-09.
2. SUSPECT IDENTITY: An automated SMS dated 2014-05-20 addresses the account holder by first name as "Peter", corroborating the case identifier "Noonien/Peter".
3. MICHIGAN CONTACT: The primary SMS counterparty +1 (248) 929-4645 uses a Southfield-area code, consistent with the GPS evidence.
4. ACTIVE COMMUNICATIONS: SMS activity on 2014-05-20 and 1,500 call log entries on 2014-05-21 confirm the device was in active use approximately ten months before forensic acquisition.
5. OUT-OF-STATE TIES: Contacts and call logs reveal ties to Vermont (Hilltop Inn, 802 area code), Massachusetts (two personal contacts with 781/978 area codes), and Connecticut (outgoing call to 860-759-2990). These indicate the suspect's network extends across the US Northeast in addition to the Michigan physical-location evidence.
6. TRAVEL-RELATED APP SEARCHES: Recovered Google Play Store search history contains three travel-related queries: "marriott mobile app" (2013-02-13), "fly delta app" (2013-06-11), and "marriott" (2014-01-15). The first two precede the GPS-confirmed Southfield trip of August 2013; the third follows it. This pattern is consistent with use of Marriott hotels and Delta Air Lines during the period of suspect activity, though it reflects only three data points out of 47 recovered searches and should not be overstated.
7. EVIDENCE INTEGRITY: MD5 and SHA1 hashes verified; chain of custody maintained throughout.

Correlated findings and conclusions from the mobile-forensics report.

9. Skills Demonstrated

Mobile forensics, FTK Imager, Autopsy, E01/E02, hash verification, Android artifacts, GPS, SQLite artifacts, communications analysis, timeline reconstruction.

10. Recommendations and Future Improvements

- Create a fully anonymized public version and add artifact IDs for every screenshot.
- Export a CSV timeline and include map visualizations.

11. Portfolio and Resume Presentation

Mobile Device Digital Forensics Investigation - An academic mobile-forensics case study examining an E01/E02 MicroSD image with FTK Imager and Autopsy. The project verified evidence integrity and correlated GPS, Maps history, SMS, call logs, contacts, installed applications, and search history into a documented timeline.

Recommended repository contents: README.md, Documentation/, Evidence/, Source/ or Original_Report/, and a clear authorization/privacy notice.

12. Completion Statement

This project is considered complete for portfolio use because the scope, implementation process, testing evidence, results, skills, limitations, and future improvements are documented. Before public release, sensitive data must be reviewed and removed.